Who Wrote This Response?

A CNN and LSTM study of LLM response attribution

Team leader: Brayden Aubry
Team member: Brayden Aubry (individual project)
Report date: October 11, 2026

Research questions

RQ1: Can response text identify the model family? **RQ2:** Does the prompt help? **RQ4:** What response style differences can we measure? We did not run RQ3 because filtering left only one source category.

1. Data and collection

We used the public Ai2 Dolci DPO Model Response Pool. It contains responses from several model configurations to shared prompts. We selected one configuration from each of three families: Gemma 3 4B, OLMo 2 7B, and Qwen 3 no-reasoning 4B. The data builder matched records by `custom_id` and kept only prompts and source labels identical across all three configurations.

To reduce use of user-originated or potentially sensitive content, the collected subset keeps only rows labeled `IF_sft_data_verified_permissive`. It excludes WildChat and DaringAnteater source rows and drops rows with simple email, phone-number, and redacted-name patterns. This is a basic screening step, not a complete privacy audit. The source dataset is ODC-BY; the dataset card also advises checking each model's output terms before redistribution.

Dataset schema

Required field	Meaning	Example
LLM_name	Model-family label	Gemma / OLMo / Qwen
LLM_Input	Prompt given to the model	User instruction
LLM_output	Model-generated response	Assistant answer

Sample balance

Family	Model configuration	Examples
Gemma	gemma3-4b	600
OLMo	olmo2-7b	600
Qwen	qwen3-no_reasoning-4b	600

The final CSV contains 1,800 rows from 600 matched prompt IDs and 1 source category. The data is saved locally rather than included in the code repository.

2. Models and training

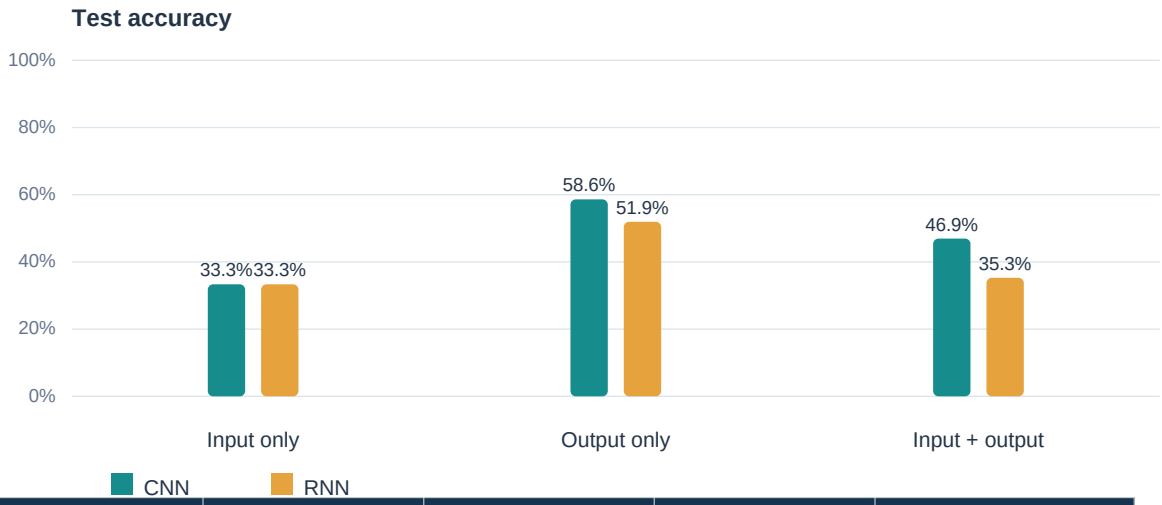
Text was lowercased and split into word and punctuation tokens. The vocabulary was learned from training data only, with a 12,000 token cap. Sequences were limited to 128 tokens. For combined text, the encoder reserved 64 tokens for the prompt and 64 for the response.

Component	CNN	RNN (LSTM)
Embedding	64 dimensions	64 dimensions
Main layer	96 filters, kernel size 3	Bidirectional LSTM, 64 hidden units
Pooling / output	Global max pooling	Concatenate final directions
Regularization	Dropout 0.3	Dropout 0.3
Optimization	Adam, learning rate 0.001	Adam, learning rate 0.001

Prompt IDs were split as 64% train, 16% validation, and 20% test. All three model responses to one prompt stayed in the same split. The test set has 120 prompt IDs and 360 rows, with 120 examples per class. Training used four epochs, batch size 64, seed 42, and the best validation-loss checkpoint. We report accuracy and macro-F1. The 3-class chance reference is 33.3% accuracy.

3. Results

Table 2 compares the two required architectures across all three input settings. The output-only CNN is the strongest required model. Both prompt-only models are at chance, as expected because each test prompt appears once for every family.



Input view	CNN accuracy	CNN macro-F1	LSTM accuracy	LSTM macro-F1
Input only	33.3%	32.7%	33.3%	33.1%
Output only	58.6%	58.6%	51.9%	51.8%
Input + output	46.9%	42.8%	35.3%	35.3%

Output-only confusion matrices

Rows are true families; columns are predicted families. The CNN correctly classified 77 Gemma, 71 OLMo, and 63 Qwen test responses. The LSTM was more likely to confuse OLMo and Qwen.

CNN	Gemma	OLMo	Qwen
Gemma	77	23	20
OLMo	18	71	31
Qwen	22	35	63

RNN	Gemma	OLMo	Qwen
Gemma	63	30	27
OLMo	9	76	35
Qwen	24	48	48

The combined view improved over prompt-only CNN and LSTM, but remained below output-only performance. The fixed 128-token budget split between prompt and response reduces the response context in that comparison.

4. Style analysis and discussion

RQ4 was explored with four simple response measurements. Average length and formatting differed across the families. These summaries are descriptive; they do not establish that a particular feature caused the CNN or LSTM predictions.

Family	Words	Lexical diversity	Punctuation rate	Lines
Gemma	354.2	0.594	0.090	29.3
OLMo	223.7	0.627	0.085	19.1
Qwen	381.2	0.587	0.082	35.1

Qwen responses were longest on average (381 words) and had the highest mean line count (35.1). OLMo responses were shortest (224 words) and had the highest lexical-diversity ratio (0.627). Gemma responses had the highest punctuation rate (0.090). The output-only CNN was the best classifier, which is consistent with response length, vocabulary, and formatting carrying some signal. A feature-removal experiment would be needed to attribute the performance to any one signal.

5. Scope, limitations, and lessons

RQ1 and RQ2: Completed with both required neural models and all three input views. The prompt-only result is a useful control: with identical prompts grouped across splits, the prompt does not explain the model label. The response itself carries the measured signal, though best accuracy is moderate at 58.6%.

RQ3: Not run. The privacy and source-license filter left one safe source category, so this dataset cannot support a valid cross-source holdout. RQ3 is extra credit in the assignment, not a required research question. The source category is not itself a verified semantic task label.

Limitations: Results cover one configuration from each family, 600 prompts per family, one random seed, and one train/validation/test split. The personal-information checks are heuristic, not a complete manual review. The 128-token cap truncates long texts. Model-family attribution here should not be generalized to every prompt, version, or deployment of these model families.

Lessons: Balanced examples and prompt-grouped splitting matter for fair comparison. Input-only performance at chance helps rule out prompt leakage. The CNN outperformed the LSTM on response-only text in this run, but repeated seeds and feature ablations are needed for stronger conclusions.

References and data use

[1] Ai2, *Dolci DPO Model Response Pool*, public dataset card. Open dataset card. Accessed October 11, 2026. Dataset license: ODC-BY; check the model-specific output terms before redistribution.

[2] Team Olmo et al. (2025), *Olmo 3*, technical report, arXiv:2512.13961.